

GrounUp Enterprise Specification

Phase 13 - Business Intelligence, Executive Analytics & Decision Intelligence

Version 1.0.0 | Review-ready | Proposed lock: GES-P13-v1.0.0

Mission

Create a governed, auditable intelligence layer that turns cross-enterprise construction data into certified metrics, operational insight, forecasts, scenarios and human-approved decisions.

Architecture Principles

Metric consistency; tenant isolation; source lineage; freshness visibility; least privilege; reproducibility; explainable AI assistance; human approval for consequential actions; versioned semantic models; measurable decision outcomes.

Cross-Phase Integration

Consumes governed data from enterprise data, estimating, projects, CRM, workforce, finance, procurement and fleet; publishes intelligence to executive and operational experiences.

Functional Domains

Executive Command Center

Enterprise scorecards, strategic objectives, portfolio health, exception management and executive narratives.

Semantic Metrics Layer

Governed KPI definitions, dimensions, measures, calendars, units, ownership and certification.

Operational Intelligence

Production, schedule, field, labor, equipment and resource analytics.

Financial Intelligence

Revenue, margin, cash, WIP, cost, forecast and profitability analytics.

Project Intelligence

Project health, cost/schedule variance, earned value, risk and forecast-at-completion.

Estimating Intelligence

Estimate quality, bid performance, production assumptions, historical calibration and estimate-to-actual.

CRM & Market Intelligence

Pipeline, win rate, customer profitability, segmentation, market and backlog analytics.

Workforce Intelligence

Utilization, productivity, overtime, qualifications, retention and workforce capacity.

Procurement Intelligence

Supplier performance, lead times, price trends, inventory and supply-chain risk.

Fleet Intelligence

Utilization, idle time, fuel, maintenance, downtime, owning/operating cost and replacement analytics.

Safety, Quality & Compliance Analytics

Leading/lagging indicators, incidents, observations, quality trends and compliance status.

Portfolio & Program Intelligence

Cross-project portfolio performance, capital allocation, backlog and strategic capacity.

Forecasting & Scenario Planning

Driver-based forecasts, what-if analysis, Monte Carlo-ready scenarios and sensitivity analysis.

AI Decision Intelligence

Anomaly detection, root-cause assistance, recommendations, confidence, evidence and human approval.

Data Platform & Lineage

Warehouse/lakehouse ingestion, transformations, lineage, freshness, quality and observability.

Reporting & Distribution

Scheduled reports, subscriptions, alerts, exports, board packs and governed sharing.

Self-Service Analytics

Role-based exploration, saved views, drill-through, natural-language query and certified datasets.

Enterprise Digital Twin

Cross-phase enterprise state, snapshots, event lineage, scenario state and decision records.

Controlled Artifact Summary

Artifact	Count
Requirements	300
Entities	123
Fields	200
Business Rules	55
Validation Rules	55
Workflows	28
Engines	35
Api Resources	45
Ui Surfaces	25
Reports	25
Kpis	40
Tests	300

KPI Registry

P13-KPI-001 - Revenue - Executive Command Center - Governed formula defined in semantic metric registry for Revenue.

P13-KPI-002 - Gross Margin % - Semantic Metrics Layer - Governed formula defined in semantic metric registry for Gross Margin %.

P13-KPI-003 - EBITDA - Operational Intelligence - Governed formula defined in semantic metric registry for EBITDA.

P13-KPI-004 - Operating Cash - Financial Intelligence - Governed formula defined in semantic metric registry for Operating Cash.

P13-KPI-005 - Cash Forecast Accuracy - Project Intelligence - Governed formula defined in semantic metric registry for Cash Forecast Accuracy.

P13-KPI-006 - WIP Over/Under Billing - Estimating Intelligence - Governed formula defined in semantic metric registry for WIP Over/Under Billing.

P13-KPI-007 - Backlog - CRM & Market Intelligence - Governed formula defined in semantic metric registry for Backlog.

P13-KPI-008 - Pipeline Coverage - Workforce Intelligence - Governed formula defined in semantic metric registry for Pipeline Coverage.

P13-KPI-009 - Bid Win Rate - Procurement Intelligence - Governed formula defined in semantic metric registry for Bid Win Rate.

P13-KPI-010 - Estimate Accuracy - Fleet Intelligence - Governed formula defined in semantic metric registry for Estimate Accuracy.

P13-KPI-011 - Project Cost Variance - Safety, Quality & Compliance Analytics - Governed formula defined in semantic metric registry for Project Cost Variance.

P13-KPI-012 - Project Schedule Variance - Portfolio & Program Intelligence - Governed formula defined in semantic metric registry for Project Schedule Variance.

P13-KPI-013 - Forecast at Completion - Forecasting & Scenario Planning - Governed formula defined in semantic metric registry for Forecast at Completion.

P13-KPI-014 - Production Plan Attainment - AI Decision Intelligence - Governed formula defined in semantic metric registry for Production Plan Attainment.

P13-KPI-015 - Labor Productivity - Data Platform & Lineage - Governed formula defined in semantic metric registry for Labor Productivity.

P13-KPI-016 - Workforce Utilization - Reporting & Distribution - Governed formula defined in semantic metric registry for Workforce Utilization.

P13-KPI-017 - Overtime % - Self-Service Analytics - Governed formula defined in semantic metric registry for Overtime %.

P13-KPI-018 - Employee Retention - Enterprise Digital Twin - Governed formula defined in semantic metric registry for Employee Retention.

P13-KPI-019 - Fleet Utilization - Executive Command Center - Governed formula defined in semantic metric registry for Fleet Utilization.

P13-KPI-020 - Idle % - Semantic Metrics Layer - Governed formula defined in semantic metric registry for Idle %.

P13-KPI-021 - Fuel Efficiency - Operational Intelligence - Governed formula defined in semantic metric registry for Fuel Efficiency.

P13-KPI-022 - Equipment Downtime - Financial Intelligence - Governed formula defined in semantic metric registry for Equipment Downtime.

P13-KPI-023 - Maintenance Cost / Hour - Project Intelligence - Governed formula defined in semantic metric registry for Maintenance Cost / Hour.

P13-KPI-024 - Procurement Savings - Estimating Intelligence - Governed formula defined in semantic metric registry for Procurement Savings.

P13-KPI-025 - Supplier On-Time Delivery - CRM & Market Intelligence - Governed formula defined in semantic metric registry for Supplier On-Time Delivery.

P13-KPI-026 - Material Lead Time - Workforce Intelligence - Governed formula defined in semantic metric registry for Material Lead Time.

P13-KPI-027 - Inventory Turnover - Procurement Intelligence - Governed formula defined in semantic metric registry for Inventory Turnover.

P13-KPI-028 - Safety TRIR - Fleet Intelligence - Governed formula defined in semantic metric registry for Safety TRIR.

P13-KPI-029 - Near-Miss Closure - Safety, Quality & Compliance Analytics - Governed formula defined in semantic metric registry for Near-Miss Closure.

P13-KPI-030 - Quality Rework % - Portfolio & Program Intelligence - Governed formula defined in semantic metric registry for Quality Rework %.

P13-KPI-031 - Customer Profitability - Forecasting & Scenario Planning - Governed formula defined in semantic metric registry for Customer Profitability.

P13-KPI-032 - Customer Retention - AI Decision Intelligence - Governed formula defined in semantic metric registry for Customer Retention.

P13-KPI-033 - Data Freshness SLA - Data Platform & Lineage - Governed formula defined in semantic metric registry for Data Freshness SLA.

P13-KPI-034 - Data Quality Pass Rate - Reporting & Distribution - Governed formula defined in semantic metric registry for Data Quality Pass Rate.

P13-KPI-035 - Forecast Accuracy - Self-Service Analytics - Governed formula defined in semantic metric registry for Forecast Accuracy.

P13-KPI-036 - AI Recommendation Acceptance - Enterprise Digital Twin - Governed formula defined in semantic metric registry for AI Recommendation Acceptance.

P13-KPI-037 - AI Recommendation Benefit - Executive Command Center - Governed formula defined in semantic metric registry for AI Recommendation Benefit.

P13-KPI-038 - Dashboard Adoption - Semantic Metrics Layer - Governed formula defined in semantic metric registry for Dashboard Adoption.

P13-KPI-039 - Decision Cycle Time - Operational Intelligence - Governed formula defined in semantic metric registry for Decision Cycle Time.

P13-KPI-040 - Strategic Objective Attainment - Financial Intelligence - Governed formula defined in semantic metric registry for Strategic Objective Attainment.

Engine Registry

P13-ENG-001 - Semantic Metrics Engine - Deterministic - Provides semantic metrics engine with governed inputs, reproducible outputs, audit trail and human controls.

P13-ENG-002 - KPI Calculation Engine - Deterministic - Provides kpi calculation engine with governed inputs, reproducible outputs, audit trail and human controls.

P13-ENG-003 - Scorecard Engine - Deterministic - Provides scorecard engine with governed inputs, reproducible outputs, audit trail and human controls.

P13-ENG-004 - Data Ingestion Engine - Deterministic - Provides data ingestion engine with governed inputs, reproducible outputs, audit trail and human controls.

P13-ENG-005 - Data Quality Engine - Deterministic - Provides data quality engine with governed inputs, reproducible outputs, audit trail and human controls.

P13-ENG-006 - Lineage Engine - Deterministic - Provides lineage engine with governed inputs, reproducible outputs, audit trail and human controls.

P13-ENG-007 - Freshness Monitor - Deterministic - Provides freshness monitor with governed inputs, reproducible outputs, audit trail and human controls.

P13-ENG-008 - Entitlement Engine - Deterministic - Provides entitlement engine with governed inputs, reproducible outputs, audit trail and human controls.

P13-ENG-009 - Dashboard Query Engine - Deterministic - Provides dashboard query engine with governed inputs, reproducible outputs, audit trail and human controls.

P13-ENG-010 - Report Distribution Engine - Deterministic - Provides report distribution engine with governed inputs, reproducible outputs, audit trail and human controls.

P13-ENG-011 - Alerting Engine - Deterministic - Provides alerting engine with governed inputs, reproducible outputs, audit trail and human controls.

P13-ENG-012 - Anomaly Detection Engine - AI-assisted - Provides anomaly detection engine with governed inputs, reproducible outputs, audit trail and human controls.

P13-ENG-013 - Forecasting Engine - AI-assisted - Provides forecasting engine with governed inputs, reproducible outputs, audit trail and human controls.

P13-ENG-014 - Scenario Engine - AI-assisted - Provides scenario engine with governed inputs, reproducible outputs, audit trail and human controls.

P13-ENG-015 - Sensitivity Engine - AI-assisted - Provides sensitivity engine with governed inputs, reproducible outputs, audit trail and human controls.

P13-ENG-016 - Root Cause Assistant - AI-assisted - Provides root cause assistant with governed inputs, reproducible outputs, audit trail and human controls.

P13-ENG-017 - Executive Narrative Engine - AI-assisted - Provides executive narrative engine with governed inputs, reproducible outputs, audit trail and human controls.

P13-ENG-018 - Recommendation Engine - AI-assisted - Provides recommendation engine with governed inputs, reproducible outputs, audit trail and human controls.

P13-ENG-019 - Confidence Engine - AI-assisted - Provides confidence engine with governed inputs, reproducible outputs, audit trail and human controls.

P13-ENG-020 - Evidence Retrieval Engine - AI-assisted - Provides evidence retrieval engine with governed inputs, reproducible outputs, audit trail and human controls.

P13-ENG-021 - Portfolio Health Engine - AI-assisted - Provides portfolio health engine with governed inputs, reproducible outputs, audit trail and human controls.

P13-ENG-022 - Project Health Engine - AI-assisted - Provides project health engine with governed inputs, reproducible outputs, audit trail and human controls.

P13-ENG-023 - Financial Intelligence Engine - AI-assisted - Provides financial intelligence engine with governed inputs, reproducible outputs, audit trail and human controls.

P13-ENG-024 - Workforce Intelligence Engine - AI-assisted - Provides workforce intelligence engine with governed inputs, reproducible outputs, audit trail and human controls.

P13-ENG-025 - Fleet Intelligence Engine - AI-assisted - Provides fleet intelligence engine with governed inputs, reproducible outputs, audit trail and human controls.

P13-ENG-026 - Procurement Intelligence Engine - AI-assisted - Provides procurement intelligence engine with governed inputs, reproducible outputs, audit trail and human controls.

P13-ENG-027 - CRM Intelligence Engine - AI-assisted - Provides crm intelligence engine with governed inputs, reproducible outputs, audit trail and human controls.

P13-ENG-028 - Estimating Intelligence Engine - AI-assisted - Provides estimating intelligence engine with governed inputs, reproducible outputs, audit trail and human controls.

P13-ENG-029 - Safety Intelligence Engine - AI-assisted - Provides safety intelligence engine with governed inputs, reproducible outputs, audit trail and human controls.

P13-ENG-030 - Quality Intelligence Engine - AI-assisted - Provides quality intelligence engine with governed inputs, reproducible outputs, audit trail and human controls.

P13-ENG-031 - Benchmarking Engine - AI-assisted - Provides benchmarking engine with governed inputs, reproducible outputs, audit trail and human controls.

P13-ENG-032 - Forecast Accuracy Engine - AI-assisted - Provides forecast accuracy engine with governed inputs, reproducible outputs, audit trail and human controls.

P13-ENG-033 - Adoption Analytics Engine - AI-assisted - Provides adoption analytics engine with governed inputs, reproducible outputs, audit trail and human controls.

P13-ENG-034 - Decision Outcome Engine - AI-assisted - Provides decision outcome engine with governed inputs, reproducible outputs, audit trail and human controls.

P13-ENG-035 - Enterprise Digital Twin Engine - AI-assisted - Provides enterprise digital twin engine with governed inputs, reproducible outputs, audit trail and human controls.

Workflow Registry

P13-WF-001 - Metric Definition & Certification - Executive Command Center - Draft > Validate > Review > Approve > Publish/Execute > Monitor > Close

P13-WF-002 - Data Source Onboarding - Semantic Metrics Layer - Draft > Validate > Review > Approve > Publish/Execute > Monitor > Close

P13-WF-003 - Ingestion & Quality Gate - Operational Intelligence - Draft > Validate > Review > Approve > Publish/Execute > Monitor > Close

P13-WF-004 - Semantic Model Publication - Financial Intelligence - Draft > Validate > Review > Approve > Publish/Execute > Monitor > Close

P13-WF-005 - Dashboard Publication - Project Intelligence - Draft > Validate > Review > Approve > Publish/Execute > Monitor > Close

P13-WF-006 - Executive Scorecard Review - Estimating Intelligence - Draft > Validate > Review > Approve > Publish/Execute > Monitor > Close

P13-WF-007 - Alert Investigation - CRM & Market Intelligence - Draft > Validate > Review > Approve > Publish/Execute > Monitor > Close

P13-WF-008 - Insight Review & Approval - Workforce Intelligence - Draft > Validate > Review > Approve > Publish/Execute > Monitor > Close

P13-WF-009 - Forecast Production - Procurement Intelligence - Draft > Validate > Review > Approve > Publish/Execute > Monitor > Close

P13-WF-010 - Scenario Planning - Fleet Intelligence - Draft > Validate > Review > Approve > Publish/Execute > Monitor > Close

P13-WF-011 - Root Cause Analysis - Safety, Quality & Compliance Analytics - Draft > Validate > Review > Approve > Publish/Execute > Monitor > Close

P13-WF-012 - Board Pack Production - Portfolio & Program Intelligence - Draft > Validate > Review > Approve > Publish/Execute > Monitor > Close

P13-WF-013 - Report Subscription - Forecasting & Scenario Planning - Draft > Validate > Review > Approve > Publish/Execute > Monitor > Close

P13-WF-014 - Self-Service Dataset Access - AI Decision Intelligence - Draft > Validate > Review > Approve > Publish/Execute > Monitor > Close

P13-WF-015 - Data Quality Incident - Data Platform & Lineage - Draft > Validate > Review > Approve > Publish/Execute > Monitor > Close

P13-WF-016 - Lineage Impact Analysis - Reporting & Distribution - Draft > Validate > Review > Approve > Publish/Execute > Monitor > Close

P13-WF-017 - Cross-Phase Reconciliation - Self-Service Analytics - Draft > Validate > Review > Approve > Publish/Execute > Monitor > Close

P13-WF-018 - AI Recommendation Review - Enterprise Digital Twin - Draft > Validate > Review > Approve > Publish/Execute > Monitor > Close

P13-WF-019 - Decision Capture - Executive Command Center - Draft > Validate > Review > Approve > Publish/Execute > Monitor > Close

P13-WF-020 - Benefit Realization Review - Semantic Metrics Layer - Draft > Validate > Review > Approve > Publish/Execute > Monitor > Close

P13-WF-021 - Portfolio Health Review - Operational Intelligence - Draft > Validate > Review > Approve > Publish/Execute > Monitor > Close

P13-WF-022 - Project Health Review - Financial Intelligence - Draft > Validate > Review > Approve > Publish/Execute > Monitor > Close

P13-WF-023 - Estimate-to-Actual Calibration - Project Intelligence - Draft > Validate > Review > Approve > Publish/Execute > Monitor > Close

P13-WF-024 - Enterprise Digital Twin Refresh - Estimating Intelligence - Draft > Validate > Review > Approve > Publish/Execute > Monitor > Close

P13-WF-025 - Executive Daily Brief - CRM & Market Intelligence - Draft > Validate > Review > Approve > Publish/Execute > Monitor > Close

P13-WF-026 - Quarterly Strategy Review - Workforce Intelligence - Draft > Validate > Review > Approve > Publish/Execute > Monitor > Close

P13-WF-027 - KPI Threshold Change - Procurement Intelligence - Draft > Validate > Review > Approve > Publish/Execute > Monitor > Close

P13-WF-028 - Analytics Release & Rollback - Fleet Intelligence - Draft > Validate > Review > Approve > Publish/Execute > Monitor > Close

Governance & Acceptance

All metrics require certified definitions, lineage and freshness. AI recommendations require evidence/confidence and human review for consequential decisions. The phase is not locked until data governance, security/privacy, AI governance, integration, UAT and configuration-management approvals are complete.